Visesh Bentula

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SUMMARY

Graduate Computer Science student specializing in **backend** and **ML-enabled systems,** with hands-on experience building and operating production-grade **microservices.** Proven impact improving **API performance, database efficiency,** and **CI/CD reliability,** with experience deploying, testing, and monitoring cloud-native services at scale**.**

EDUCATION

Texas A&M University *San Antonio, USA*
MS Computer Science; GPA: 3.78/4.0 *January 2025 – Expected May 2026*

Coursework: Data structures and Algorithms, AI, Machine Learning & Deep Learning, Operating Systems, Secure Software Development.

EXPERIENCE

Software Engineer *January 2024 – December 2024*
Buoyant Software Solutions *Hyderabad, India*

- Designed and delivered core ERP backend modules integrating CADMATIC-generated engineering data with relational databases, supporting workflows across **16 enterprise departments.** Optimized schema design, indexing, and query patterns to **reduce average SQL query latency from ~8s to ~350ms,** improving system throughput for data-intensive operations.
- Owned **production readiness and deployment workflows**, implementing Dockerized CI/CD pipelines using GitHub Actions with automated testing, rollback support, and release validation to enable **zero-downtime deployments**. Partnered with frontend and QA teams to improve end-user performance, reducing **page load times by ~45%** (Lighthouse, API profiling).
- Strengthened system reliability by introducing **unit and integration testing, structured logging, and health checks**, contributing to **zero Sev-1 production incidents over a 6-month period** and supporting safe, continuous delivery across multiple teams.

Software Developer August 2023 – January 2024
Syra Health (Syra Health Corp.) Hyderabad, India

- Developed **cloud-based backend services** for a healthcare platform, enabling real-time data ingestion, search, and monitoring in a **high-availability, security-sensitive environment**. Focusing on API efficiency and database access patterns, reducing **average request latency by ~25%** through query tuning and endpoint optimization.
- Automated build, test, and deployment workflows to reduce release cycles by ~40%, and built analytics dashboards exposing usage trends and system metrics, supporting product decisions and driving ~20% higher user engagement.

Software Development Intern *January 2023 – April 2023*
RADCube (RADcube LLC) *Hyderabad, India*

- Implemented Java-based backend components using JDBC and built MERN-stack prototypes integrating REST APIs and structured data, gaining hands-on experience with object-oriented design and collaborative full-stack.

PROJECTS

Context-Aware AI Assistant (GitHub + Slack + Postgres)

Python, Model Context Protocol (MCP), Large Language Models (LLMs), FastAPI, Postgres, GitHub API, Slack API, matplotlib

- **Engineered an AI assistant** using **MCP and LLMs** to integrate **GitHub, Slack, and Postgres,** exposing structured tool invocation via **FastAPI** endpoints to enable natural language queries across code repositories, team discussions, and system metrics.
- Implemented automated summarization and **security guardrails** including repository allowlists, read-only tool scopes, and PII filtering, reducing manual search time by **~60%** while enforcing safe, compliant access to enterprise data sources.

Flood Semantic Segmentation (U-Net)

Python, PyTorch (encoder-decoder CNN), TorchVision, scikit-learn, NumPy, CUDA/GPU acceleration, FastAPI

- Developed a custom **U-Net semantic segmentation model** on the **FloodNet aerial imagery dataset**, achieving **~93%** validation pixel accuracy and **0.73 mIoU**, outperforming a majority-class baseline by **>60%** in accuracy and **>4×** in mIoU. Exported the model to **ONNX** and deployed a FastAPI inference service, measuring GPU and CPU latency for batch and single-image workloads.
- Built a **GPU-accelerated training pipeline** with optimized data loaders and metric computation, delivering **~0.89** macro precision and **~0.90** macro recall, and performed confusion-matrix-driven error analysis to assess robustness across flood and non-flood regions.

SKILLS

- **Languages:** Python, Java, SQL, JavaScript, C
- **Data & Infrastructure:** PostgreSQL, MySQL, MongoDB, Redis, Docker, AWS/GCP, Kafka
- **Backend & Systems:** REST APIs, Microservices, Authentication & RBAC, Data Modeling.
- **Frontend Technologies:** React, Angular, Typescript